

Life Cycle Assessment Report

Product: Test Botanical Drink 250ml
Company: Fixture Beverages Ltd
Functional unit: 1 unit of Test Botanical Drink 250ml as sold
System boundary: Cradle-to-Grave
Reference year: 2025 Report date: 15 May 2026 Version: 1.0
Prepared in accordance with ISO 14040:2006, ISO 14044:2006 and ISO 14067:2018.
Generated by: FixtureLCA Platform

1. Goal and Scope (ISO 14044 4.2)

Intended application: quantify the product carbon footprint for sustainability disclosures.
Intended audience: internal stakeholders, B2B customers, consumers.
Comparative assertion: No. This study does not support public comparative assertions.
Included stages: raw materials, processing, packaging, distribution, use phase, end of life.
Excluded: capital goods, employee commuting.
Cut-off criteria: mass <1%, energy <1%; any flow contributing >1% to any impact category is included regardless of mass or energy contribution.

2. Methodology (ISO 14044 4.3, 4.4)

LCIA method: ReCiPe 2016 v1.1 Midpoint (H). Climate change: IPCC AR6 GWP-100 (2021):
CO2 = 1, CH4 fossil = 29.8, CH4 biogenic = 27.0, N2O = 273 kg CO2e/kg.
Databases: ecoinvent 3.12 (11 factors), AGRIBALYSE 3.2 (5 factors), DEFRA 2025 (4 factors).
Allocation: Economic allocation (market price) applied to all co-products. Physical allocation was feasible but not used; no justification for

3. Data Quality Assessment (ISO 14044 4.2.3.6)

Overall data quality: 62% (Fair), impact-weighted average of material confidence scores.
Pedigree matrix scores (1 best - 5 worst): reliability 3, completeness 2, temporal 3, geographic 2, technological 2.
Primary data: 0% (0 of 12 materials). Secondary data: 100% (12 materials). Proxy: 0%.
Material-level quality: Apple Juice (secondary, HIGH, 85%, Kent UK);
Elderflower Extract (secondary, HIGH, 85%, Surrey UK); Sugar (secondary, MEDIUM, 50%, Norfolk UK);
Water (secondary, MEDIUM, 50%, Reading UK); Glass 250ml Bottle (secondary, MEDIUM, 50%, Rotherham UK);
Aluminium Cap (secondary, MEDIUM, 50%, Cranleigh UK); Label (secondary, MEDIUM, 50%, Hungerford UK).

4. Climate Impact

Total climate impact (GWP-100): 0.5200 kg CO2e per functional unit.
Lifecycle stage breakdown (kg CO2e):

Raw Materials	0.1800	(34.6%)
Processing	0.0600	(11.5%)
Packaging	0.2600	(50.0%)
Distribution	0.0300	(5.8%)
Use Phase	0.0100	(1.9%)
End of Life	-0.0200	(-3.8%)

Stage total: 0.5200 kg CO2e.
Scope split: Scope 1 0%, Scope 2 0%, Scope 3 100% (contract manufacture).

5. Detailed GHG Reporting (ISO 14067 6.4.9.3)

GHG species breakdown (mass kg, CO2e kg, GWP-100):

CO2 (fossil)	0.4400	0.4400	1
CO2 (biogenic)	0.0600	0.0600	1
CO2 (LULUC)	0.0000	0.0000	1
CH4 (fossil)	0.0005	0.0149	29.8

CH4 (biogenic) 0.0000 0.0000 27.0

N2O 0.0000186 0.0051 273

Total GWP-100 (all species): 0.5200 kg CO2e.

Fossil carbon footprint (excl. biogenic): 0.4600 kg CO2e.

Biogenic CO2 is reported separately from fossil CO2 per ISO 14067:2018 6.4.9.3.

6. End of Life

Gross EoL emissions: 0.0300 kg CO2e. Recycling credits (avoided burden): -0.0500 kg CO2e.

Net EoL impact: -0.0200 kg CO2e.

Disposal pathways: Glass 250ml Bottle (74% recycling, 8% landfill, 18% incineration);

Aluminium Cap (72% recycling, 8% landfill, 20% incineration);

Label (74% recycling, 5% landfill, 15% incineration, 4% composting, 2% AD).

7. Interpretation, Uncertainty and Review (ISO 14044 4.5, 6)

Significant issues: packaging dominates the footprint (50.0%); glass bottle is the largest single contributor.

Uncertainty: propagated uncertainty +/-24% (95% CI, 0.395 - 0.645 kg CO2e), root-sum-of-squares with pedigree-derived GSDs. Sensitivity: +/-20% variation on top 3 contributors changes the total by up to +/-9%.

Critical review: NOT CONDUCTED. A critical review per ISO 14044 Clause 6 is recommended before public comparative assertions.

Recommendations: collect primary data for the glass bottle; verify transport distances.